ChE 205 – Fall 2014, Spring 2015, Fall 2015, Spring 2016
Computational Methods in Chemical Engineering
Assessment results for final exam
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PI	Description
1	Apply mathematical and scientific principles to formulate models and systems relevant to the subject
2	Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra

Student Learning Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Fall 2014

Number of Students: 30

SO	1	2
% (3 = Satisfactory)	63%	63%
% (2 = Developing)	20%	20%
% (1 = Unsatisfactory)	17%	17%
Average Score	2.13	2.13
Standard Deviation	0.87	0.87

Spring 2015

Number of Students: 62

SO	1	2
% (3 = Satisfactory)	63%	63%
% (2 = Developing)	26%	26%
% (1 = Unsatisfactory)	11%	11%
Average Score	2.20	2.20
Standard Deviation	0.75	0.75

Fall 2015

Number of Students: 42

SO	1	2
% (3 = Satisfactory)	67%	67%
% (2 = Developing)	21%	21%
% (1 = Unsatisfactory)	12%	12%
Average Score	2.30	2.30
Standard Deviation	0.56	0.56

Fall 2016

Number of Students: 36

SO	1	2
% (3 = Satisfactory)	56%	56%
% (2 = Developing)	19%	19%
% (1 = Unsatisfactory)	25%	25%
Average Score	1.75	1.75
Standard Deviation	1.12	1.12

Conclusions:

Results of this analysis indicated that the percentage of students scoring at the satisfactory level ranged from 56% to 67% and the students scoring at the unsatisfactory level ranged from 11% to 25%. It is noted that the Fall 2016 scores were lower than previous years.

ChE 205 – Fall 2014
Computational Methods in Chemical Engineering
Assessment Rubric for Final Exam
(Problem 4) Solving ODE for 1D heat transfer in a pipe using the Shooting Method

Level Criterion	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
SO (1), (2) Quality of Calculations	Calculations/algorithms are incorrect or presented in a confusing manner	Calculations/algorithms are partly correct and presented with errors in explanation/execution	Calculations/algorithms are mostly correct and presented with minor errors in explanation/execution

Mapping of grading rubric to assessment rubric:

Rubric item #1 and 2: Problem 4 of final exam (0-22 point scale adjusted to 1-3 scale).
Scoring scale: ≥ 2.0 (Satisfactory); 1.9—1.0 (Developing); < 1.0 (Unsatisfactory)

Example Work:

Problem 4 (22 Points): The change in temperature of a fluid flowing through a section of heated pipe of length L is given by the following:

$$\alpha \frac{d^2T}{dx^2} = v\frac{dT}{dx} - k(T_w - T) \tag{1}$$

where T is the temperature; α is the thermal diffusivity of the fluid, v is the average fluid velocity, k is the heat transfer coefficient; T_W is the temperature of the pipe wall. The temperature entering the pipe is $T(x = 0) = T_{in}$.

a) (5 points) Write equation (1) as two-first order ordinary differential equations (ODEs).
b) (10 points) Apply the forward Euler method to the ODEs obtained in part (a), and write the equations in the explicit form. **For example if dT/dx = f(T, x), solve for T_{i+1}.**
c) (4 points) Write the boundary conditions at x = 0 and x = L.
d) (3 points) Describe how you would solve this problem using the shooting method in Excel.

a) $\boxed{\frac{dT}{dx} = F}$ +2 $\quad \frac{d^2T}{dx^2} = \frac{dF}{dx} = \frac{V}{\alpha}\frac{dT}{dx} - k(T_w - T) \Rightarrow \boxed{\frac{dF}{dx} = \frac{V}{\alpha}F - \frac{k}{\alpha}(T_w - T)}$ +3

b) $\frac{T_{i+1} - T_i}{\Delta x} = F_i$ +2 $\Rightarrow \boxed{T_{i+1} = T_i + \Delta x F_i}$ +2

$\frac{F_{i+1} - F_i}{\Delta x}$ +2 $= \frac{V}{\alpha}F_i$ +1 $- \frac{k}{\alpha}(T_w - T_i)$ +1 $\Rightarrow F_{i+1} = F_i + \frac{\Delta x V}{\alpha}F_i - \frac{\Delta x k}{\alpha}(T_w - T_i)$

$\boxed{F_{i+1} = F_i\left(1 + \frac{\Delta x V}{\alpha}\right) - \frac{\Delta x k}{\alpha}(T_w - T_i)}$ +2

c) $T(x=0) = T_{in}$ +1 $\qquad F(x=0) = ?$ +1 $\qquad F(x=L) = 0$ +2

d) Solve for $F(x=0)$ while satisfying BC $F(x=L)=0$

use Excel Solver or Goal Seek +3